Supplementary Material to: Decoding Pre-Response EEG Signatures of Choice, Accuracy, and Confidence in Perceptual Decision-Making

All numbers are generated from results_final.json by the scripts in the reproducibility package. Table numbers S1 –S12 are referenced from the manuscript.

Table S1. File-system times of the plan, scripts, and result files (the only available evidence of order; no external time stamp, plan edited in place)

File	Content	Created	Last modified
research_paper_draft.pdf	Original unfinished draft (no results)	2026-10-13 09:02	2026-10-13 09:02
analysis/extract.py	Epoch extraction script	2026-10-13 09:16	2026-10-13 09:16
analysis/ ANALYSIS_PLAN.md	Internal analysis plan (last edit = second amendment)	2026-10-13 09:20	2026-10-13 18:13
analysis/run_main.py	Main analysis script	2026-10-13 09:21	2026-10-13 09:40
analysis/results.json	Main results (first submitted manuscript)	2026-10-13 09:40	2026-10-13 14:13
analysis/rq4_extra.py	Sequential interaction analysis script	2026-10-13 17:05	2026-10-13 17:05
analysis/ results_rq4_interaction.json	Sequential interaction results	2026-10-13 17:05	2026-10-13 17:05
analysis/dir_control.py	Stimulus-direction control script	2026-10-13 17:08	2026-10-13 17:08
analysis/ results_dir_control.json	Stimulus-direction results	2026-10-13 17:08	2026-10-13 17:08
analysis/extra_analyses.py	Time-resolved and RT-tercile script	2026-10-13 17:55	2026-10-13 17:55
analysis/results_extra.json	Time-resolved and RT-tercile results	2026-10-13 17:56	2026-10-13 17:56
analysis/extra2.py	Calibration-definition sensitivity script (300 permutations)	2026-10-13 17:59	2026-10-13 17:59
analysis/results_extra2.json	Calibration-definition sensitivity results	2026-10-13 18:00	2026-10-13 18:00
analysis/compare_decoders.py	Comparison-decoder script (second amendment)	2026-10-13 18:13	2026-10-13 18:13
analysis/results_compare.json	Comparison-decoder results	2026-10-13 18:19	2026-10-13 18:19
journal_submission/ Revised_Manuscript.docx	Manuscript submitted to the journal	2026-10-13 17:54	2026-10-14 20:30

Table S2a. Retention (percent and number of trials) for every window of the grid; the selected window is L = 600 ms, B = 150 ms

L \ B (ms)	0	50	100	150	200	250	300	350
200	100.0% (7377)	100.0% (7377)	100.0% (7377)	100.0% (7377)	99.9% (7370)	99.5% (7341)	98.3% (7249)	95.9% (7071)
300	100.0% (7377)	100.0% (7377)	99.9% (7370)	99.5% (7341)	98.3% (7249)	95.9% (7071)	92.9% (6851)	89.7% (6615)
400	99.9% (7370)	99.5% (7341)	98.3% (7249)	95.9% (7071)	92.9% (6851)	89.7% (6615)	86.2% (6358)	82.6% (6090)
500	98.3% (7249)	95.9% (7071)	92.9% (6851)	89.7% (6615)	86.2% (6358)	82.6% (6090)	78.0% (5756)	73.5% (5423)
600	92.9% (6851)	89.7% (6615)	86.2% (6358)	82.6% (6090)	78.0% (5756)	73.5% (5423)	68.6% (5059)	62.4% (4600)
700	86.2% (6358)	82.6% (6090)	78.0% (5756)	73.5% (5423)	68.6% (5059)	62.4% (4600)	54.8% (4041)	46.0% (3394)
800	78.0% (5756)	73.5% (5423)	68.6% (5059)	62.4% (4600)	54.8% (4041)	46.0% (3394)	37.9% (2797)	29.1% (2149)

Table S2b. Mean held-out-participant AUC for choice over the window grid (descriptive; not used for selection; – = retention below 50%)

L \ B (ms)	0	50	100	150	200	250	300	350
200	0.527	0.518	0.511	0.524	0.525	0.526	0.519	0.510
300	0.522	0.523	0.522	0.527	0.521	0.511	0.513	0.519
400	0.531	0.527	0.520	0.519	0.513	0.522	0.521	0.520
500	0.528	0.516	0.507	0.517	0.522	0.519	0.506	0.499
600	0.519	0.518	0.515	0.527	0.514	0.507	0.503	0.505
700	0.529	0.529	0.519	0.511	0.505	0.526	0.494	–
800	0.542	0.520	0.504	0.525	0.510	–	–	–

Table S2b. Mean held-out-participant AUC for accuracy over the window grid (descriptive; not used for selection; – = retention below 50%)

L \ B (ms)	0	50	100	150	200	250	300	350
200	0.513	0.511	0.508	0.517	0.526	0.540	0.526	0.531
300	0.521	0.522	0.514	0.534	0.546	0.547	0.543	0.538
400	0.529	0.540	0.541	0.546	0.554	0.552	0.543	0.540
500	0.554	0.551	0.553	0.555	0.557	0.544	0.526	0.546
600	0.557	0.555	0.557	0.535	0.533	0.548	0.533	0.512
700	0.561	0.546	0.541	0.541	0.533	0.510	0.521	–
800	0.546	0.542	0.511	0.492	0.507	–	–	–

Table S2b. Mean held-out-participant AUC for confidence over the window grid (descriptive; not used for selection; – = retention below 50%)

L \ B (ms)	0	50	100	150	200	250	300	350
200	0.530	0.527	0.522	0.522	0.504	0.527	0.533	0.534
300	0.533	0.533	0.521	0.521	0.514	0.550	0.544	0.554
400	0.518	0.522	0.526	0.537	0.526	0.563	0.566	0.561
500	0.535	0.545	0.539	0.555	0.547	0.565	0.569	0.542
600	0.538	0.545	0.549	0.540	0.544	0.529	0.512	0.504
700	0.556	0.550	0.541	0.529	0.479	0.484	0.502	–
800	0.561	0.523	0.498	0.457	0.483	–	–	–

Table S3. Per-participant trial counts, retention, and out-of-participant AUC (primary decoder)

Participant	Valid	Retained	Retention	Choice AUC	Accuracy AUC	Confidence AUC	Incorr./high
1	317	179	56%	0.530	0.553	0.578	2
2	301	300	100%	0.470	0.542	0.524	22
3	312	300	96%	0.491	0.497	0.459	2
4	318	116	36%	0.511	0.530	0.462	0
5	306	298	97%	0.513	0.607	0.559	10
6	313	268	86%	0.533	0.484	0.537	10
7	306	225	74%	0.564	0.552	0.519	5
8	314	314	100%	0.474	0.556	0.474	1
9	290	265	91%	0.564	0.544	0.445	26
10	309	309	100%	0.527	0.592	0.617	17
11	318	86	27%	0.536	0.455	0.594	4
12	309	299	97%	0.540	0.533	0.607	17
13	301	244	81%	0.580	0.470	0.681	6
14	300	300	100%	0.521	0.543	0.546	11
15	307	307	100%	0.517	0.470	0.451	1
16	297	296	100%	0.540	0.565	0.592	7
17	303	89	29%	0.506	0.521	0.494	1
18	291	284	98%	0.476	0.555	0.606	43
19	301	299	99%	0.616	0.531	0.474	6

20	302	302	100%	0.526	0.491	0.560	8
21	318	317	100%	0.486	0.512	0.461	6
22	318	77	24%	0.624	0.485	0.544	9
23	310	309	100%	0.503	0.609	0.638	4
24	316	307	97%	0.491	0.639	0.548	19

Table S4a. Comparison decoders: mean held-out-participant AUC on raw and flexibly RT-residualized features

Decoder (features| model)	Spec	Choice raw	Choice resid	Acc. raw	Acc. resid	Conf. raw	Conf. resid	Non-conv.
amp|lr		0.527	0.527	0.535	0.502	0.540	0.504	0
amp|lda		0.525	0.526	0.536	0.503	0.541	0.505	0
amp|svm_lin		0.527	0.527	0.535	0.502	0.541	0.504	0
amp|svm_rbf		0.510	0.520	0.532	0.504	0.550	0.502	0
amp|rf		0.511	0.504	0.533	0.495	0.551	0.505	0
amp|mlp		0.517	0.514	0.522	0.497	0.532	0.495	0
amp4|lr		0.515	0.518	0.541	0.513	0.560	0.517	0
amp4|lda		0.514	0.519	0.542	0.512	0.563	0.515	0
logvar|lr		0.512	0.508	0.476	0.472	0.471	0.485	0
amp+logvar| lr		0.530	0.528	0.510	0.481	0.506	0.488	0

Table S4b. Decoder specifications (scikit-learn 1.0.2 unless stated; primary decoder = exact Newton solver for the same objective)

Model	Specification
lr	FastLR{'C': 0.1, 'class_weight': 'balanced', 'solver': 'exact Newton (gradient tol 1e-9)', 'penalty': 'l2', 'intercept_penalized': False}
lda	LinearDiscriminantAnalysis{'covariance_estimator': None, 'n_components': None, 'priors': None, 'shrinkage': 'auto', 'solver': 'lsqr', 'store_covariance': False, 'tol': 0.0001}
svm_lin	LinearSVC{'C': 0.01, 'class_weight': 'balanced', 'dual': False, 'fit_intercept': True, 'intercept_scaling': 1, 'loss': 'squared_hinge', 'max_iter': 1000, 'multi_class': 'ovr', 'penalty': 'l2', 'random_state': None, 'tol': 0.0001, 'verbose': 0}
svm_rbf	SVC{'C': 1.0, 'break_ties': False, 'cache_size': 200, 'class_weight': 'balanced', 'coef0': 0.0, 'decision_function_shape': 'ovr', 'degree': 3, 'gamma': 'scale', 'kernel': 'rbf', 'max_iter': -1, 'probability': False, 'random_state': None, 'shrinking': True, 'tol': 0.001, 'verbose': False}
rf	RandomForestClassifier{'bootstrap': True, 'ccp_alpha': 0.0, 'class_weight': 'balanced_subsample', 'criterion': 'gini', 'max_depth': None, 'max_features': 'auto', 'max_leaf_nodes': None, 'max_samples': None, 'min_impurity_decrease': 0.0, 'min_samples_leaf': 20, 'min_samples_split': 2, 'min_weight_fraction_leaf': 0.0, 'n_estimators': 300, 'n_jobs': 1, 'oob_score': False, 'random_state': 0, 'verbose': 0, 'warm_start': False}
mlp	MLPClassifier{'activation': 'relu', 'alpha': 0.1, 'batch_size': 'auto', 'beta_1': 0.9, 'beta_2': 0.999, 'early_stopping': True, 'epsilon': 1e-08, 'hidden_layer_sizes': (32,), 'learning_rate': 'constant', 'learning_rate_init': 0.001, 'max_fun': 15000, 'max_iter': 200, 'momentum': 0.9, 'n_iter_no_change': 10, 'nesterovs_momentum': True, 'power_t': 0.5, 'random_state': 0, 'shuffle': True, 'solver': 'adam', 'tol': 0.0001, 'validation_fraction': 0.1, 'verbose': False, 'warm_start': False}

Table S4c. Differences from the primary decoder with equivalence verdicts

Target / variant	Decoder	Diff. to primary	90% CI	Wilcoxon p	Holm p	Verdict (margin ±0.02)

choice/raw	amp|lda	-0.0012	[-0.002, -0.000]	0.037	0.330	equivalent (90 percent CI within +-0.02)
choice/raw	amp|svm_lin	-0.0001	[-0.000, +0.000]	0.546	1.000	equivalent (90 percent CI within +-0.02)
choice/raw	amp|svm_rbf	-0.0164	[-0.029, -0.005]	0.049	0.392	inconclusive
choice/raw	amp|rf	-0.0158	[-0.028, -0.004]	0.101	0.708	inconclusive
choice/raw	amp|mlp	-0.0100	[-0.021, +0.001]	0.317	1.000	inconclusive
choice/raw	amp4|lr	-0.0120	[-0.026, +0.000]	0.390	1.000	inconclusive
choice/raw	amp4|lda	-0.0124	[-0.026, +0.000]	0.345	1.000	inconclusive
choice/raw	logvar|lr	-0.0146	[-0.040, +0.011]	0.331	1.000	inconclusive
choice/raw	amp+logvar|lr	+0.0030	[-0.012, +0.019]	0.790	1.000	equivalent (90 percent CI within +-0.02)
accuracy/raw	amp|lda	+0.0014	[-0.000, +0.003]	0.412	1.000	equivalent (90 percent CI within +-0.02)
accuracy/raw	amp|svm_lin	+0.0003	[-0.000, +0.001]	0.060	0.423	equivalent (90 percent CI within +-0.02)
accuracy/raw	amp|svm_rbf	-0.0028	[-0.013, +0.008]	0.565	1.000	equivalent (90 percent CI within +-0.02)
accuracy/raw	amp|rf	-0.0019	[-0.012, +0.008]	0.768	1.000	equivalent (90 percent CI within +-0.02)
accuracy/raw	amp|mlp	-0.0129	[-0.030, +0.006]	0.089	0.536	inconclusive
accuracy/raw	amp4|lr	+0.0060	[-0.006, +0.018]	0.360	1.000	equivalent (90 percent CI within +-0.02)
accuracy/raw	amp4|lda	+0.0067	[-0.006, +0.019]	0.331	1.000	equivalent (90 percent CI within +-0.02)
accuracy/raw	logvar|lr	-0.0593	[-0.082, -0.038]	0.000	0.002	worse (CI below -margin)
accuracy/raw	amp+logvar|lr	-0.0246	[-0.036, -0.014]	0.004	0.031	inconclusive
confidence/raw	amp|lda	+0.0002	[-0.002, +0.002]	0.858	1.000	equivalent (90 percent CI within +-0.02)
confidence/raw	amp|svm_lin	+0.0006	[-0.000, +0.001]	0.140	0.977	equivalent (90 percent CI within +-0.02)
confidence/raw	amp|svm_rbf	+0.0094	[-0.009, +0.028]	0.390	1.000	inconclusive
confidence/raw	amp|rf	+0.0106	[-0.009, +0.032]	0.603	1.000	inconclusive
confidence/raw	amp|mlp	-0.0086	[-0.019, +0.002]	0.153	0.977	equivalent (90 percent CI within +-0.02)
confidence/raw	amp4|lr	+0.0192	[-0.004, +0.044]	0.360	1.000	inconclusive
confidence/raw	amp4|lda	+0.0223	[-0.001, +0.048]	0.208	1.000	inconclusive
confidence/raw	logvar|lr	-0.0694	[-0.104, -0.035]	0.001	0.013	worse (CI below -margin)
confidence/raw	amp+logvar|lr	-0.0348	[-0.054, -0.017]	0.002	0.018	inconclusive
choice/resid	amp|lda	-0.0005	[-0.002, +0.001]	0.761	1.000	equivalent (90 percent CI within +-0.02)
choice/resid	amp|svm_lin	-0.0001	[-0.000, +0.000]	1.000	1.000	equivalent (90 percent CI within +-0.02)
choice/resid	amp|svm_rbf	-0.0065	[-0.020, +0.006]	0.966	1.000	inconclusive
choice/resid	amp|rf	-0.0224	[-0.040, -0.006]	0.015	0.135	inconclusive
choice/resid	amp|mlp	-0.0130	[-0.028, +0.002]	0.331	1.000	inconclusive
choice/resid	amp4|lr	-0.0087	[-0.022, +0.004]	0.509	1.000	inconclusive
choice/resid	amp4|lda	-0.0083	[-0.022, +0.004]	0.584	1.000	inconclusive
choice/resid	logvar|lr	-0.0185	[-0.044, +0.007]	0.264	1.000	inconclusive
choice/resid	amp+logvar|lr	+0.0014	[-0.013, +0.016]	0.922	1.000	equivalent (90 percent CI within +-0.02)
accuracy/resid	amp|lda	+0.0011	[+0.000, +0.002]	0.169	1.000	equivalent (90 percent CI within +-0.02)

accuracy/resid	amp|svm_lin	+0.0001	[-0.000, +0.000]	0.693	1.000	equivalent (90 percent CI within +-0.02)
accuracy/resid	amp|svm_rbf	+0.0020	[-0.010, +0.014]	0.790	1.000	equivalent (90 percent CI within +-0.02)
accuracy/resid	amp|rf	-0.0064	[-0.022, +0.010]	0.252	1.000	inconclusive
accuracy/resid	amp|mlp	-0.0045	[-0.023, +0.015]	0.509	1.000	inconclusive
accuracy/resid	amp4|lr	+0.0113	[-0.003, +0.026]	0.345	1.000	inconclusive
accuracy/resid	amp4|lda	+0.0101	[-0.004, +0.025]	0.473	1.000	inconclusive
accuracy/resid	logvar|lr	-0.0300	[-0.056, -0.004]	0.069	0.553	inconclusive
accuracy/resid	amp+logvar|lr	-0.0204	[-0.033, -0.008]	0.019	0.175	inconclusive
confidence/resid	amp|lda	+0.0003	[-0.001, +0.002]	0.903	1.000	equivalent (90 percent CI within +-0.02)
confidence/resid	amp|svm_lin	+0.0000	[-0.001, +0.001]	0.876	1.000	equivalent (90 percent CI within +-0.02)
confidence/resid	amp|svm_rbf	-0.0019	[-0.020, +0.016]	0.989	1.000	inconclusive
confidence/resid	amp|rf	+0.0003	[-0.020, +0.022]	1.000	1.000	inconclusive
confidence/resid	amp|mlp	-0.0090	[-0.028, +0.011]	0.375	1.000	inconclusive
confidence/resid	amp4|lr	+0.0124	[-0.009, +0.034]	0.491	1.000	inconclusive
confidence/resid	amp4|lda	+0.0110	[-0.011, +0.033]	0.584	1.000	inconclusive
confidence/resid	logvar|lr	-0.0194	[-0.043, +0.006]	0.229	1.000	inconclusive
confidence/resid	amp+logvar|lr	-0.0158	[-0.031, -0.001]	0.101	0.910	inconclusive

Table S4d. Diagnostics for the below-chance log-variance decoder (500 permutations of the whole LOSO procedure; orientation: positive class = correct / high confidence)

Target	Observed mean AUC	Null mean	Null 5th pct	Lower-tail p	Participants < 0.5	Within-participant 10-fold AUC	Pattern corr. held-out vs train	Frac. negative	Training AUC (full model)
accuracy	0.476	0.500	0.481	0.024	16/24	0.506	-0.101	0.79	0.554
confidence	0.471	0.500	0.471	0.048	15/24	0.542	-0.032	0.58	0.582

Table S5. Validation of the exact Newton solver against scikit-learn's L2 logistic regression (lbfgs, balanced class weights, C = 0.1)

Target	Mean AUC (Newton)	Mean AUC (scikit-learn)	|difference|	Max participant | difference|	Score correlation	Max |coefficient difference|
choice	0.52665	0.52665	1.03e-06	6.75e-04	0.9999998	1.90e-04
accuracy	0.53486	0.53487	1.06e-05	1.66e-04	0.9999999	1.38e-04
confidence	0.54041	0.54054	1.36e-04	1.54e-03	0.9999999	1.15e-04

Table S6. Class counts per held-out participant for the contrast IH vs IL (evaluable = at least 3 trials per class)

Participant	Positive-class trials	Negative-class trials	Evaluable
1	2	20	no
2	22	28	yes
3	2	52	no
4	0	10	no
5	10	25	yes
6	10	13	yes
7	5	33	yes
8	1	50	no
9	26	7	yes

10	17	50	yes
11	4	4	no
12	17	25	yes
13	6	40	yes
14	11	12	yes
15	1	39	no
16	7	21	yes
17	1	12	no
18	43	17	yes
19	6	52	yes
20	8	22	yes
21	6	13	yes
22	9	17	yes
23	4	34	yes
24	19	20	yes

Table S6. Class counts per held-out participant for the contrast IH vs CH (evaluable = at least 3 trials per class)

Participant	Positive-class trials	Negative-class trials	Evaluable
1	2	34	no
2	22	139	yes
3	2	36	no
4	0	31	no
5	10	111	yes
6	10	94	yes
7	5	46	yes
8	1	12	no
9	26	71	yes
10	17	55	yes
11	4	28	yes
12	17	147	yes
13	6	42	yes
14	11	111	yes
15	1	21	no
16	7	110	yes
17	1	26	no
18	43	119	yes
19	6	36	yes
20	8	144	yes
21	6	62	yes
22	9	12	yes
23	4	37	yes
24	19	105	yes

Table S6. Class counts per held-out participant for the contrast IL vs CL (evaluable = at least 3 trials per class)

Participant	Positive-class trials	Negative-class trials	Evaluable
1	20	15	yes
2	28	21	yes
3	52	117	yes
4	10	8	yes
5	25	20	yes
6	13	25	yes
7	33	52	yes
8	50	63	yes
9	7	12	yes

10	50	63	yes
11	4	3	no
12	25	17	yes
13	40	35	yes
14	12	23	yes
15	39	44	yes
16	21	46	yes
17	12	12	yes
18	17	24	yes
19	52	95	yes
20	22	32	yes
21	13	16	yes
22	17	18	yes
23	34	46	yes
24	20	32	yes

Table S6. Class counts per held-out participant for the contrast CH vs CL (evaluable = at least 3 trials per class)

Participant	Positive-class trials	Negative-class trials	Evaluable
1	34	15	yes
2	139	21	yes
3	36	117	yes
4	31	8	yes
5	111	20	yes
6	94	25	yes
7	46	52	yes
8	12	63	yes
9	71	12	yes
10	55	63	yes
11	28	3	yes
12	147	17	yes
13	42	35	yes
14	111	23	yes
15	21	44	yes
16	110	46	yes
17	26	12	yes
18	119	24	yes
19	36	95	yes
20	144	32	yes
21	62	16	yes
22	12	18	yes
23	37	46	yes
24	105	32	yes

Table S7. Sequential GEE models (5332 trial pairs, 24 participants; formulas in Section 4.4; S = neural state, C = reported confidence, Sf = state from the RT-residualized decoder)

Model	Term	β	SE robust	p robust	SE bias-reduced	p (t, 23 df)	95% CI (bias-reduced)	Odds ratio [95% CI]
M1	S	0.0192	0.0368	0.602	0.0387	0.625	[-0.061, 0.099]	1.02 [0.94, 1.10]
M1	D	1.1529	0.1231	0.000	0.1300	0.000	[0.884, 1.422]	3.17 [2.42, 4.14]
M1	A	-0.4042	0.0802	0.000	0.0849	0.000	[-0.580, -0.229]	0.67 [0.56, 0.80]
M1	T	-0.3248	0.2446	0.184	0.2569	0.219	[-0.856, 0.207]	0.72 [0.42, 1.23]

M1c	C	0.1430	0.0475	0.003	0.0503	0.009	[0.039, 0.247]	1.15 [1.04, 1.28]
M1c	D	1.1550	0.1244	0.000	0.1313	0.000	[0.883, 1.427]	3.17 [2.42, 4.16]
M1c	A	-0.4849	0.0887	0.000	0.0942	0.000	[-0.680, -0.290]	0.62 [0.51, 0.75]
M1c	T	-0.1517	0.2055	0.460	0.2163	0.490	[-0.599, 0.296]	0.86 [0.55, 1.34]
M1j	S	0.0204	0.0375	0.586	0.0394	0.609	[-0.061, 0.102]	1.02 [0.94, 1.11]
M1j	C	0.1432	0.0477	0.003	0.0505	0.009	[0.039, 0.248]	1.15 [1.04, 1.28]
M1j	D	1.1547	0.1244	0.000	0.1313	0.000	[0.883, 1.426]	3.17 [2.42, 4.16]
M1j	A	-0.4863	0.0878	0.000	0.0932	0.000	[-0.679, -0.293]	0.61 [0.51, 0.75]
M1j	T	-0.1181	0.2241	0.598	0.2360	0.621	[-0.606, 0.370]	0.89 [0.55, 1.45]
M1i	S	0.0136	0.0426	0.749	0.0447	0.764	[-0.079, 0.106]	1.01 [0.92, 1.11]
M1i	D	-2.0038	0.1629	0.000	0.1730	0.000	[-2.362, -1.646]	0.13 [0.09, 0.19]
M1i	A	-2.5294	0.1545	0.000	0.1636	0.000	[-2.868, -2.191]	0.08 [0.06, 0.11]
M1i	D:A	4.3045	0.2330	0.000	0.2465	0.000	[3.795, 4.814]	74.03 [44.46, 123.26]
M1i	T	-0.6559	0.2886	0.023	0.3028	0.041	[-1.282, -0.029]	0.52 [0.28, 0.97]
M1ci	C	0.1795	0.0507	0.000	0.0538	0.003	[0.068, 0.291]	1.20 [1.07, 1.34]
M1ci	D	-2.0237	0.1634	0.000	0.1735	0.000	[-2.383, -1.665]	0.13 [0.09, 0.19]
M1ci	A	-2.6462	0.1670	0.000	0.1769	0.000	[-3.012, -2.280]	0.07 [0.05, 0.10]
M1ci	D:A	4.3356	0.2344	0.000	0.2478	0.000	[3.823, 4.848]	76.37 [45.74, 127.53]
M1ci	T	-0.4172	0.2479	0.092	0.2612	0.124	[-0.958, 0.123]	0.66 [0.38, 1.13]
M1f	Sf	0.0227	0.0300	0.449	0.0316	0.479	[-0.043, 0.088]	1.02 [0.96, 1.09]
M1f	D	1.1529	0.1230	0.000	0.1299	0.000	[0.884, 1.422]	3.17 [2.42, 4.14]
M1f	A	-0.4043	0.0802	0.000	0.0849	0.000	[-0.580, -0.229]	0.67 [0.56, 0.80]
M1f	T	-0.3576	0.2263	0.114	0.2376	0.146	[-0.849, 0.134]	0.70 [0.43, 1.14]
M2	S	0.0006	0.0021	0.766	0.0022	0.782	[-0.004, 0.005]	
M2	T	0.1720	0.0328	0.000	0.0350	0.000	[0.100, 0.244]	
M2	A	0.0061	0.0059	0.297	0.0062	0.330	[-0.007, 0.019]	
M2c	C	0.0043	0.0033	0.195	0.0035	0.235	[-0.003, 0.012]	
M2c	T	0.1775	0.0325	0.000	0.0347	0.000	[0.106, 0.249]	
M2c	A	0.0038	0.0064	0.551	0.0067	0.577	[-0.010, 0.018]	
M3	S	0.0396	0.0306	0.196	0.0323	0.233	[-0.027, 0.106]	1.04 [0.97, 1.11]
M3	A	0.1007	0.1061	0.343	0.1121	0.379	[-0.131, 0.333]	1.11 [0.88, 1.39]
M3	T	0.0428	0.3071	0.889	0.3249	0.896	[-0.629, 0.715]	1.04 [0.53, 2.04]
M3c	C	0.0182	0.0283	0.520	0.0302	0.553	[-0.044, 0.081]	1.02 [0.96, 1.08]
M3c	A	0.0929	0.1092	0.395	0.1157	0.430	[-0.147, 0.332]	1.10 [0.86, 1.39]
M3c	T	0.0036	0.3065	0.991	0.3244	0.991	[-0.667, 0.675]	1.00 [0.51, 1.96]

Table S7b. Participant bootstrap that re-fits the decoder (500 resamples)

Model	Coefficient	Point estimate	Bootstrap 95% interval	Bootstrap p (two-sided)	Replicates used / failed
M1	S	0.0192	[-0.054, 0.082]	0.721	479 / 21
M1c	C	0.1430	[0.050, 0.240]	0.008	482 / 18
M2	S	0.0006	[-0.003, 0.005]	0.830	500 / 0
M3	S	0.0396	[-0.023, 0.104]	0.287	500 / 0

Table S8. Stimulus-locked, causally available decoding (window = 300 ms ending at τ; trials with no response by τ + δ)

δ (ms)	τ (ms)	Target	Trials	Mean AUC	p (one-sided t-test)	Holm p
0	400	choice	7366	0.500	0.523	1.000
0	500	choice	7208	0.511	0.080	0.640
0	600	choice	6777	0.511	0.136	0.815
0	700	choice	6281	0.485	0.956	1.000
0	800	choice	5659	0.485	0.917	1.000
0	900	choice	4925	0.514	0.098	0.689
0	1000	choice	3845	0.501	0.485	1.000
0	1100	choice	2587	0.508	0.306	1.000
0	400	accuracy	7366	0.487	0.937	1.000
0	500	accuracy	7208	0.484	0.950	1.000
0	600	accuracy	6777	0.501	0.438	1.000
0	700	accuracy	6281	0.522	0.006	0.045
0	800	accuracy	5659	0.514	0.075	0.525
0	900	accuracy	4925	0.501	0.469	1.000
0	1000	accuracy	3845	0.504	0.344	1.000
0	1100	accuracy	2587	0.490	0.810	1.000
0	400	confidence	4140	0.491	0.819	1.000
0	500	confidence	4025	0.494	0.707	1.000
0	600	confidence	3744	0.501	0.476	1.000
0	700	confidence	3429	0.475	0.957	1.000
0	800	confidence	3079	0.485	0.815	1.000
0	900	confidence	2644	0.498	0.551	1.000
0	1000	confidence	2033	0.505	0.429	1.000
0	1100	confidence	1374	0.472	0.808	1.000
150	400	choice	7015	0.498	0.594	1.000
150	500	choice	6546	0.505	0.261	1.000
150	600	choice	6010	0.488	0.858	1.000
150	700	choice	5319	0.483	0.944	1.000
150	800	choice	4432	0.487	0.894	1.000
150	900	choice	3199	0.532	0.015	0.124
150	1000	choice	1947	0.499	0.515	1.000
150	1100	choice	962	0.490	0.681	1.000
150	400	accuracy	7015	0.486	0.948	1.000
150	500	accuracy	6546	0.489	0.870	1.000
150	600	accuracy	6010	0.498	0.600	1.000
150	700	accuracy	5319	0.524	0.027	0.220
150	800	accuracy	4432	0.500	0.502	1.000
150	900	accuracy	3199	0.508	0.294	1.000
150	1000	accuracy	1947	0.491	0.714	1.000
150	1100	accuracy	962	0.488	0.706	1.000
150	400	confidence	3900	0.492	0.721	1.000
150	500	confidence	3597	0.500	0.493	1.000
150	600	confidence	3278	0.494	0.635	1.000
150	700	confidence	2883	0.467	0.915	1.000
150	800	confidence	2367	0.500	0.500	1.000
150	900	confidence	1682	0.493	0.610	1.000
150	1000	confidence	1021	0.528	0.124	0.994
150	1100	confidence	490	0.473	0.702	1.000

Table S9a. Primary AUCs under operations that use no sample after the window end (2265 trials with all variants available)

Variant	Choice AUC (Δ [95% CI])	Accuracy AUC (Δ [95% CI])	Confidence AUC (Δ [95% CI])	Rel. RMS diff. of features	Within-participant feature correlation
V0 (zero-phase decimation of the whole recording; used throughout)	0.5186	0.5562	0.5437		
V1_boxcar	0.5192 (+0.0006 [-0.0001, +0.0013])	0.5559 (-0.0002 [-0.0016, +0.0013])	0.5437 (+0.0000 [-0.0014, +0.0014])	0.0241	0.99976
V2_causal_decim	0.5170 (-0.0016 [-0.0040, +0.0008])	0.5568 (+0.0006 [-0.0028, +0.0035])	0.5372 (-0.0065 [-0.0136, +0.0012])	0.1336	0.99087
V3_trunc_decim	0.5186 (-0.0000 [-0.0001, +0.0000])	0.5562 (+0.0000 [-0.0001, +0.0001])	0.5437 (-0.0000 [-0.0002, +0.0002])	0.0004	1.00000
P0_zerophase_full	0.5135 (-0.0051 [-0.0159, +0.0029])	0.5464 (-0.0098 [-0.0154, -0.0040])	0.5448 (+0.0011 [-0.0051, +0.0076])	0.1184	0.99184
P1_causal	0.5148 (-0.0038 [-0.0404, +0.0271])	0.4893 (-0.0669 [-0.0985, -0.0267])	0.5220 (-0.0217 [-0.0855, +0.0375])	1.3275	0.16226
P2_zerophase_trunc	0.4783 (-0.0404 [-0.0651, -0.0176])	0.4864 (-0.0697 [-0.0963, -0.0416])	0.5444 (+0.0006 [-0.0333, +0.0342])	1.1410	0.25570

Table S9b. Analytic temporal support of the operations that we control (effective kernel of 'operation, then 600 ms window mean')

Operation	Support (ms, |h| > 0.001 of peak)	Share of kernel weight after window end	…beyond 3 ms	…beyond 150 ms
zero-phase decimation filter (as used for the epochs)	112	0.245%	0.142%	0.0002%
causal decimation filter	143	0.000%	0.000%	0.0000%
proxy: zero-phase 0.5-40 Hz Butterworth band-pass (order 4)	1715	31.722%	31.678%	22.9117%
proxy: causal 0.5-40 Hz Butterworth band-pass	2946	0.000%	0.000%	0.0000%

Table S10. Normalization and RT-regression variants (mean held-out-participant AUC; differences relative to the canonical retained-trial batch normalization)

Variant	Choice	Accuracy	Confidence	Δ choice [95% CI]	Δ accuracy [95% CI]	Δ confidence [95% CI]
canonical	0.527	0.535	0.540			
all_trials_batch	0.526	0.535	0.540	-0.0003 [-0.0020, +0.0016]	+0.0001 [-0.0027, +0.0033]	-0.0003 [-0.0038, +0.0027]
training_only_pooled	0.524	0.537	0.544	-0.0031 [-0.0229, +0.0145]	+0.0023 [-0.0092, +0.0140]	+0.0031 [-0.0148, +0.0225]
separate_calibration_run	0.525	0.534	0.536	-0.0021 [-0.0068, +0.0025]	-0.0005 [-0.0057, +0.0046]	-0.0043 [-0.0137, +0.0061]
causal_running	0.513	0.544	0.540	-0.0134 [-0.0283, -0.0003]	+0.0090 [-0.0017, +0.0199]	-0.0009 [-0.0240, +0.0197]
rt_linear_training_only	0.525	0.502	0.519			
rt_flex_training_only	0.527	0.503	0.517			

Table S10b. Definitions of the variants

Variant	Definition
canonical	per-participant z-score over the participant's retained trials (target

	participant's unlabeled batch used; labels never used)
all_trials_batch	per-participant z-score over ALL valid trials of the participant, including excluded fast trials (normalization of the first submitted manuscript's Table 4)
training_only_pooled	no target-participant data used: every feature standardized with the mean/SD of the pooled training participants' raw features
separate_calibration_run	held-out participant's run r standardized with statistics of the participant's OTHER run (unlabeled calibration session); training participants as canonical
causal_running	each trial standardized with the mean/SD of the participant's earlier retained trials in the same run (>= 20 earlier trials required; trials with fewer are dropped from training and testing)
rt_linear_transductive	canonical features residualized on the participant's own z(ln RT) (needs completed RTs of the target participant)
rt_linear_training_only	residualization coefficients estimated on training participants only and applied to the held-out participant (no target RT regression)
rt_flex_training_only	as above with the 5-df natural-spline RT basis

Table S11. Observed effects, interval estimates, and minimum detectable effects (definitions in Section 4.4; permutation null percentiles are rejection thresholds, not power)

Analysis	Observed	95% interval	Minimum detectable effect (80% power)
choice: mean participant AUC	0.527	[0.511, 0.543]	0.525
accuracy: mean participant AUC	0.535	[0.517, 0.554]	0.527
confidence: mean participant AUC	0.540	[0.515, 0.566]	0.539
IH_vs_IL: pooled AUC	0.488	[0.450, 0.528]	0.557
IH_vs_CH: pooled AUC	0.531	[0.496, 0.576]	0.562
IL_vs_CL: pooled AUC	0.516	[0.492, 0.543]	0.547
CH_vs_CL: pooled AUC	0.525	[0.506, 0.549]	0.528
GEE M1:S	0.0192	[-0.061, 0.099]	b = 0.108 (OR 1.11/SD)
GEE M1c:C	0.1430	[0.039, 0.247]	b = 0.141 (OR 1.15/SD)
GEE M1j:S	0.0204	[-0.061, 0.102]	b = 0.110 (OR 1.12/SD)
GEE M1j:C	0.1432	[0.039, 0.248]	b = 0.141 (OR 1.15/SD)
GEE M1i:S	0.0136	[-0.079, 0.106]	b = 0.125 (OR 1.13/SD)
GEE M1ci:C	0.1795	[0.068, 0.291]	b = 0.151 (OR 1.16/SD)
GEE M1f:Sf	0.0227	[-0.043, 0.088]	b = 0.088 (OR 1.09/SD)
GEE M2:S	0.0006	[-0.004, 0.005]	b = 0.006
GEE M2c:C	0.0043	[-0.003, 0.012]	b = 0.010
GEE M3:S	0.0396	[-0.027, 0.106]	b = 0.090 (OR 1.09/SD)
GEE M3c:C	0.0182	[-0.044, 0.081]	b = 0.085 (OR 1.09/SD)

Table S12. Time-resolved response-locked decoding in 100 ms windows (retained trials with RT ≥ 750 ms; one-sided t-test p and Holm p over the 14 windows)

Window (ms re. response)	Choice AUC (p, Holm p)	Accuracy AUC (p, Holm p)	Confidence AUC (p, Holm p)	Accuracy resid. AUC (Holm p)
[-750, -650)	0.496 (0.659, 1.000)	0.499 (0.529, 1.000)	0.526 (0.010, 0.118)	0.496 (1.000)
[-700, -600)	0.511 (0.130, 1.000)	0.519 (0.046, 0.369)	0.535 (0.009, 0.113)	0.507 (1.000)
[-650, -550)	0.508 (0.186, 1.000)	0.529 (0.002, 0.022)	0.525 (0.050, 0.503)	0.512 (1.000)
[-600, -500)	0.519 (0.014, 0.183)	0.517 (0.003, 0.032)	0.527 (0.025, 0.278)	0.507 (1.000)
[-550, -450)	0.499 (0.529, 1.000)	0.507 (0.296, 0.887)	0.517 (0.116, 0.696)	0.485 (1.000)
[-500, -400)	0.480 (0.958, 1.000)	0.525 (0.005, 0.059)	0.527 (0.066, 0.543)	0.510 (1.000)
[-450, -350)	0.505 (0.237, 1.000)	0.527 (0.000, 0.005)	0.521 (0.062, 0.543)	0.508 (1.000)
[-400, -300)	0.514 (0.153, 1.000)	0.525 (0.006, 0.059)	0.521 (0.060, 0.543)	0.504 (1.000)
[-350, -250)	0.503 (0.390, 1.000)	0.525 (0.031, 0.276)	0.516 (0.175, 0.813)	0.496 (1.000)
[-300, -200)	0.488 (0.890, 1.000)	0.515 (0.052, 0.369)	0.514 (0.163, 0.813)	0.496 (1.000)
[-250, -150)	0.516 (0.016, 0.186)	0.511 (0.111, 0.591)	0.541 (0.004, 0.060)	0.493 (1.000)
[-200, -100)	0.506 (0.232, 1.000)	0.508 (0.209, 0.837)	0.500 (0.497, 0.995)	0.498 (1.000)
[-150, -50)	0.525 (0.011, 0.158)	0.492 (0.832, 1.000)	0.493 (0.748, 0.995)	0.497 (1.000)

[-100, 0)	0.510 (0.238, 1.000)	0.510 (0.098, 0.591)	0.510 (0.176, 0.813)	0.513 (0.384)

Additional tables and figures moved from the main text

Table S13. Closely related EEG studies

Study	Signal and timing	Main finding
{Philiastides}	Single-trial components about 170 and 300 ms after the stimulus	Components predicted accuracy; a separate component tracked difficulty
{Gherman}	Single-trial component before the choice and the confidence report (EEG-fMRI)	Stimulus-independent confidence component, linked to ventromedial prefrontal activity
{Boldt}	Pe after the response	Pe graded with confidence independent of accuracy
{Rausch}	350–500 ms after the stimulus and 250–350 ms after the response	Two confidence-related periods; post-response activity partly distinct
{Desender19b}	Time-resolved decoding from the initial decision onward	Confidence decodable; no above-chance decoding before the response
{Desender19a}	Centro-parietal component sensitive to confidence	Predicted the bound increase on the next trial
{Dou24}	Centro-parietal positivity slope before the decision, within participant	Predicted confidence with accuracy, RT, and evidence strength controlled
{Balsdon24}	Post-decision confidence weights applied to pre-response EEG, within participant	Online confidence signal preceding motor accumulation
{Fernandez21}	Stimulus- and response-locked mean amplitudes plus RT; leave-one-participant and leave-one-task-out	Continuous confidence predicted better than a mean baseline; RT hurt transfer

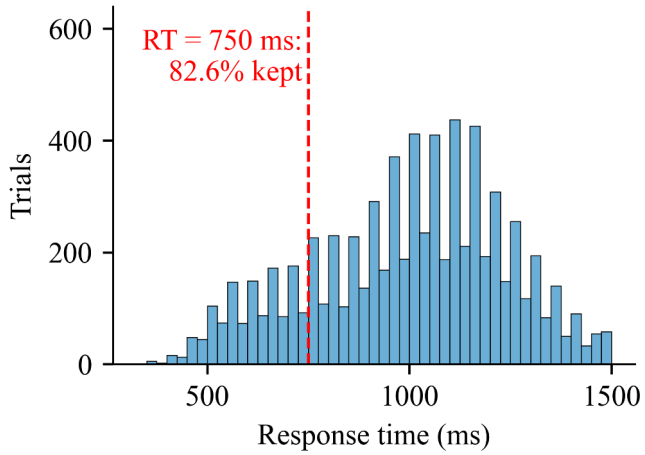


Figure S14. Distribution of response times for all valid trials. Trials left of the dashed line (RT below L + B = 750 ms) are too fast for the selected window and are excluded.

Table S15. Operations preceding the window average and their temporal support

Operation	Source	Phase / support	Uses samples after window end?
Gradient-artifact template subtraction (80 volumes)	Dataset	Templates centred on each volume	Possibly (artifact estimate)
12 ms median filter	Dataset	Presumably centred, 12 ms	Possibly, up to 6 ms
0.5–40 Hz band-pass	Dataset	Type, order, phase not stated	Not determinable
Ocular and cardiac component removal	Dataset	Spatial projection from the recordings	Not determinable
Baseline, 100 ms before	Dataset	Earlier samples only	No

stimulus			
Decimation 1000 to 250 Hz (used for epochs)	This study	Zero phase, Chebyshev I, order 8; kernel below 0.001 of peak beyond 112 ms	0.25% of kernel weight
Window average	This study	Samples inside the window	No
Feature and RT normalization	This study	Statistics of the participant's retained trials	Yes (batch; see the corresponding item)

Table S16. Accuracy by confidence trial counts in the analysis window

Cell	Trials	Participants
Correct/low	839	24
Correct/high	1,629	24
Incorrect/low	616	24
Incorrect/high	237	23

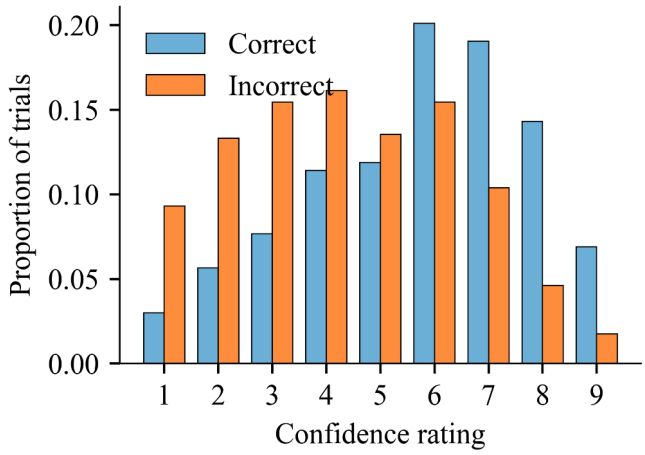


Figure S17. Distribution of confidence ratings for correct and incorrect trials (all valid trials).

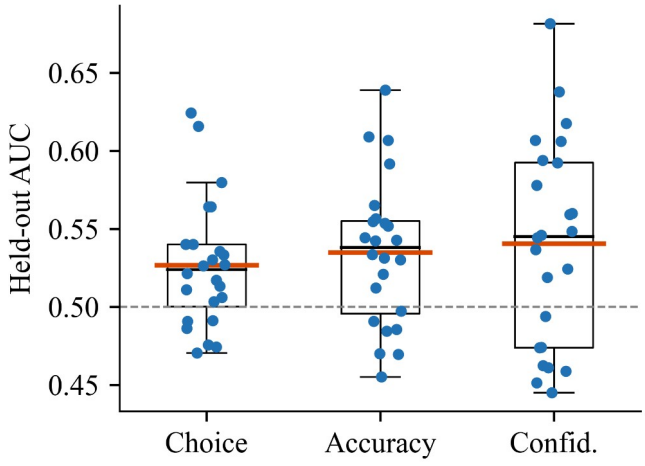


Figure S18. Held-out-participant AUC for choice, accuracy, and confidence (high versus low). Dots are participants, boxes show quartiles, black lines the medians, and orange lines the means; the dashed line is chance.

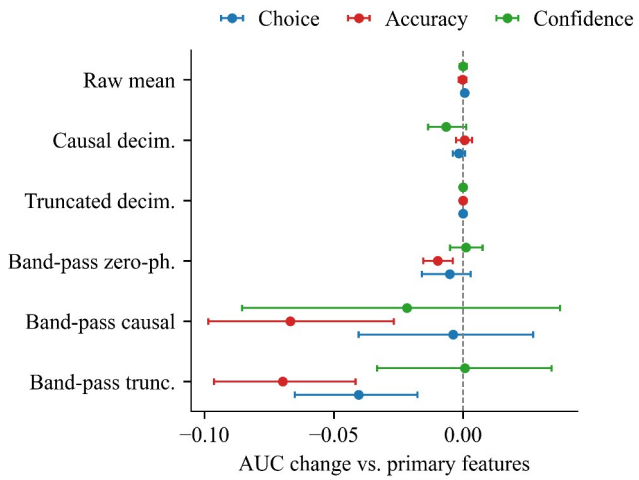


Figure S19. Change in mean AUC relative to the primary features for each variant of the temporal processing, with 95% bootstrap intervals over participants, in the subset of participants that could be re-extracted.

Table S20. Normalization and RT-regression variants (mean held-out-participant AUC)

Variant	Choice	Accuracy	Confid.
Canonical batch	0.527	0.535	0.540
All-trial batch	0.526	0.535	0.540
Training-only pooled	0.524	0.537	0.544
Calibration run	0.525	0.534	0.536
Causal running	0.513	0.544	0.540
RT lin., train-only	0.525	0.502	0.519
RT spline, train-only	0.527	0.503	0.517

Table S21. Decoder comparison, raw and RT-residualized mean AUC

Decoder	Choice	Accuracy	Confid.
Mean amplitude, logistic (primary)	0.527/.527	0.535/.502	0.540/.504
Mean amplitude, shrinkage LDA	0.525/.526*	0.536/.503*	0.541/.505*
Mean amplitude, linear SVM	0.527/.527*	0.535/.502*	0.541/.504*
Mean amplitude, RBF SVM	0.510/.520	0.532/.504*	0.550/.502
Mean amplitude, random forest	0.511/.504	0.533/.495*	0.551/.505
Mean amplitude, perceptron	0.517/.514	0.522/.497	0.532/.495*
Four sub-bin means, logistic	0.515/.518	0.541/.513*	0.560/.517
Four sub-bin means, LDA	0.514/.519	0.542/.512*	0.563/.515
Log variance, logistic	0.512/.508	0.476/.472	0.471/.485
Mean + log variance, logistic	0.530/.528*	0.510/.481	0.506/.488

Table S22. Diagnostics for the log-variance decoder

Quantity	Accuracy	Confid.
Observed mean AUC	0.476	0.471
Permutation null mean (5th pct.)	0.500 (0.481)	0.500 (0.471)
Lower-tail permutation p	0.024	0.048
Participants below 0.5	16/24	15/24
Within-participant 10-fold AUC	0.506	0.542
Pattern correlation, held-out vs. training	−0.10	−0.03
Full-model training AUC	0.554	0.582

Table S23. Component baselines and the whole-scalp decoder (mean AUC, permutation p)

Features	Choice	Accuracy	Confid.
CPP-like	0.509 (0.217)	0.508 (0.299)	0.533 (0.016)
LRP-like	0.495 (0.740)	0.489 (0.823)	0.514 (0.208)
CPP-like + LRP-like	0.503 (0.431)	0.501 (0.505)	0.529 (0.034)
RT only (flexible)	0.505	0.614	0.688
CPP-like + LRP-like + RT	0.499 (0.573)	0.613 (< 0.001)	0.688 (< 0.001)
Whole-scalp (primary)	0.527 (0.005)	0.535 (0.003)	0.540 (0.006)

Table S24. Mean AUC for selected windows of the grid (L, B in ms)

L, B	Choice	Accuracy	Confid.
400, 100	0.520	0.541	0.526
400, 150	0.519	0.546	0.537
600, 100	0.515	0.557	0.549
600, 150	0.527	0.535	0.540
600, 200	0.514	0.533	0.544
800, 100	0.504	0.511	0.498
800, 150	0.525	0.492	0.457

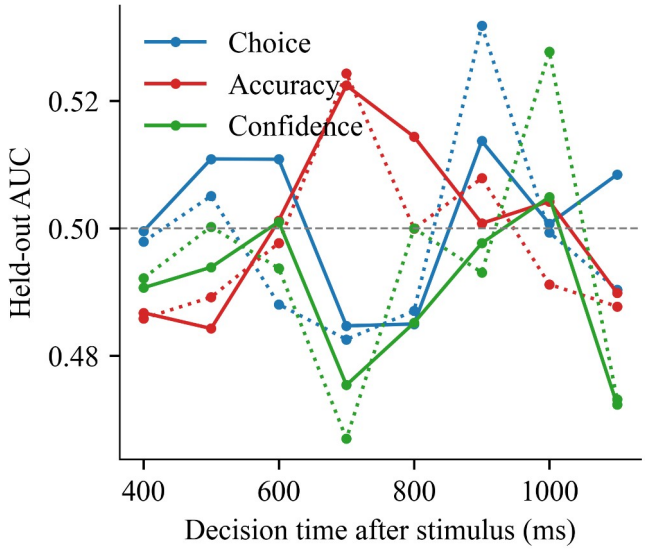


Figure S25. Stimulus-locked decoding with the window ending at the decision time after stimulus onset. Solid lines: trials with no response by that time; dotted lines: no response within 150 ms after it.

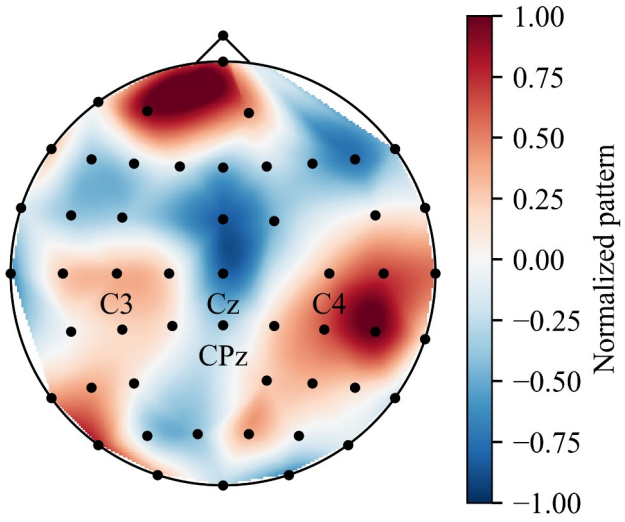


Figure S26. Activation pattern of the choice decoder (normalized), averaged over leave-one-participant-out folds; red indicates higher amplitude for a right-hand choice.

Table S27. Confidence contrasts by definition of the classes (pooled AUC, permutation p)

Split	Incorrect	Correct
Ratings 1–3 vs 7–9	0.488 (0.524)	0.525 (0.015)
Ratings 1–2 vs 8–9	0.533 (0.059)	0.548 (0.005)
Ratings 1–4 vs 6–9	0.507 (0.256)	0.524 (0.009)

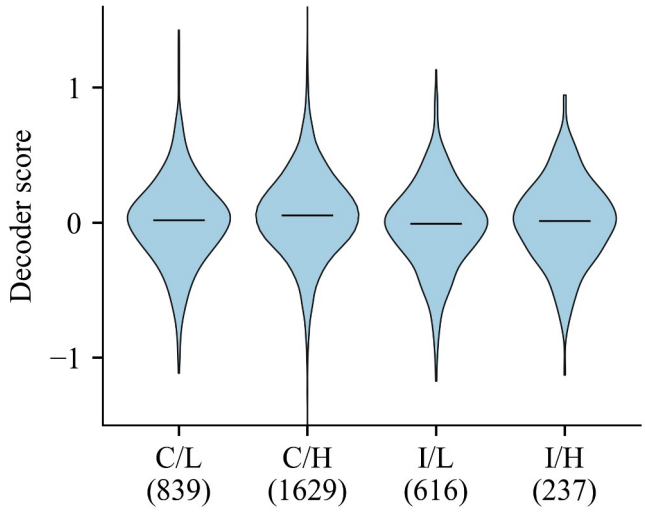


Figure S28. Out-of-participant accuracy-decoder score by calibration cell (C/I = correct/incorrect, L/H = low/high confidence; trial counts in parentheses). Higher scores are more correct-like; horizontal lines are medians.